

EGGSHELL FOR OSTEOARTHRITIS TREATMENT

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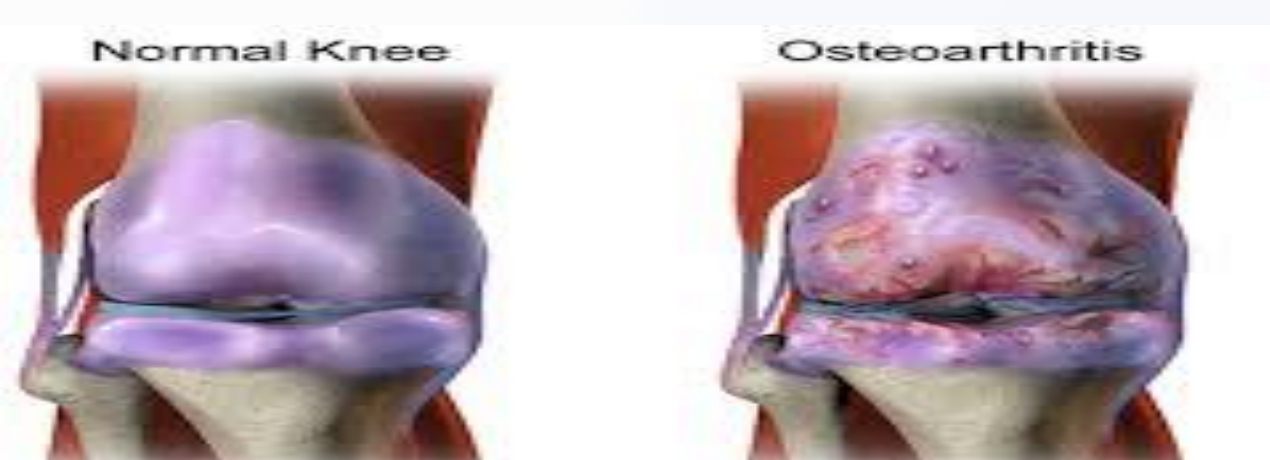
Purpose

Is there anything worse than pain? Of course, not so one of the most difficult human pains is the pain of osteoarthritis, the cartilage within a joint begins to break down and the underlying bone begins to change.

These changes usually develop slowly and get worse over time.

OA can cause pain, stiffness, and swelling. On the other hand, there is a problem represented

in the survival of eggshells as a cause of pollution and climatic changes through the production of carbon dioxide and methane during fermentation in the rubbish dump, so do you think these two problems can overcome each other?



Materials

Materials	Picture	Quantity
Eggshell membrane		1 gram
Pepsin		2 tables
phosphate buffer		40MI
Hydrogen chloride (HCl)		2MI
Arduino system		-----
Lenses		1
water		100ml

Abstract

One does not know how weak he until he is ill.....!
Egypt is one of the countries that suffer most from many health problems such as cancer, kidney and heart diseases, especially bone diseases, where most Egyptian people suffer from bone diseases such as herniated discs, osteoporosis, and osteoarthritis. After a lot of research, we found that most of them suffer from osteoarthritis, so why don't we think about relieving their pain instead of surgical operations? Sometimes the doctor resorts to injecting the patient with an injection that contains hyaluronic acid. It is expensive, so after a lot of research, we found many previous solutions, such as obtaining that acid from bovine tissue, and also obtaining the rooster's combs. But we thought about the problem of eggshells, so tons of eggshells are thrown away every day, so why not take advantage of them? Therefore, we decided to take some steps to extract hyaluronic acid from the eggshell membrane. First, we grind 1g from the eggshells membrane, then adding 4mL of HCl to the solution ,then adding of 2 tables of pepsin, then 100mL of water, then adding 40 mL of phosphate buffer and boil at 40-45 degrees Celsius, after that the finally step which is to collect the hyaluronic after precipitation.
So our project is safe, has low cost, recycles eggshells, and relieves the pain of people who suffer from osteoarthritis.

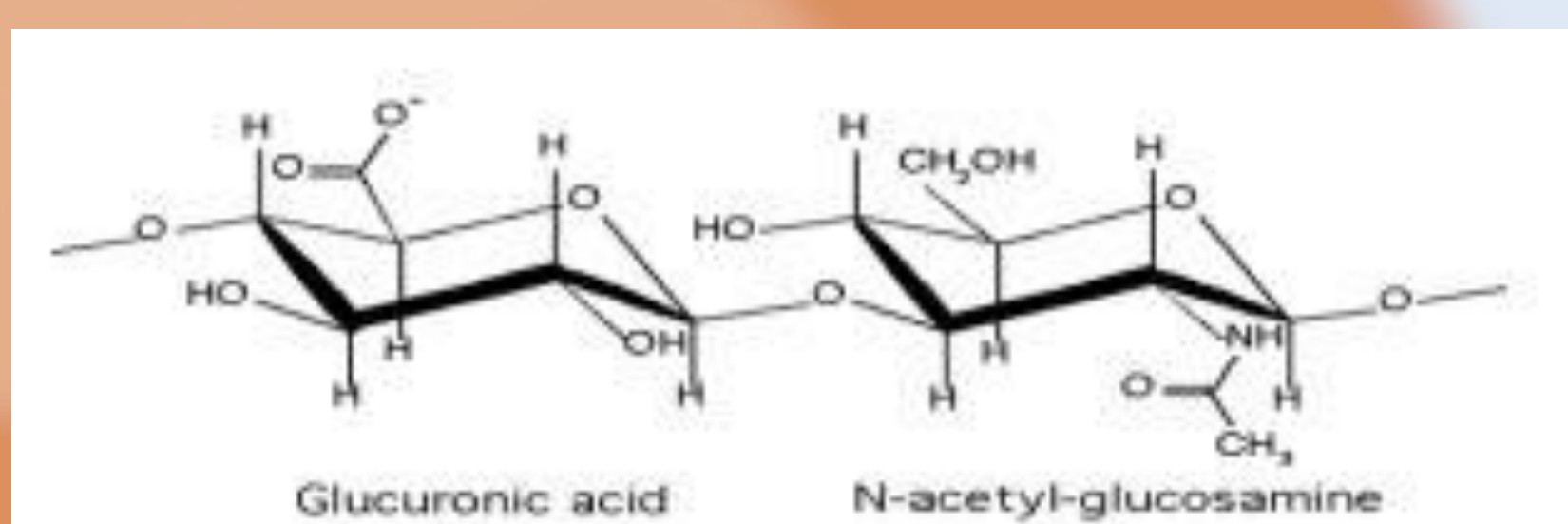
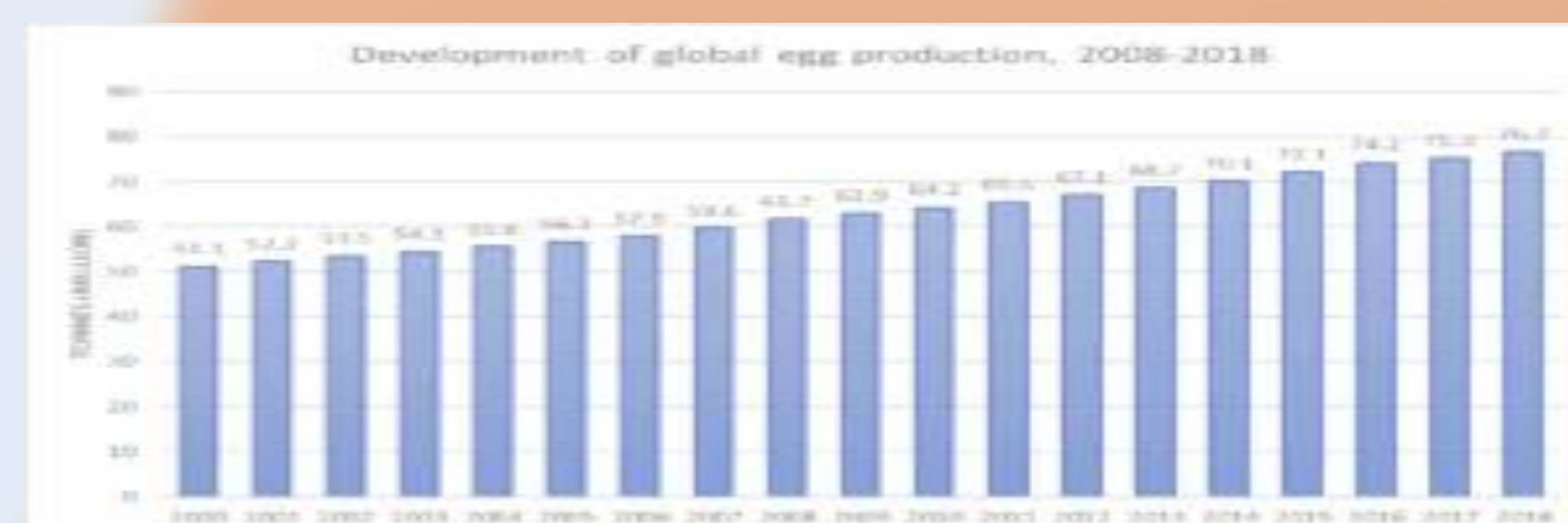
procedure

To do the experiment in successful way we follow a lot of steps which are:-

- First, removing the membrane from the eggshell by dissolves the calcium carbonate of the shell with using HCl.
- Second, drying the eggshells membrane at room temperature for one day .
- Third , dry eggshells membrane were grinded by using Manual mill.
- Fourth, adding 2 tables of pepsin in 100 mL of water and stirring this solution for 10 minutes.
- Fifth, adding 4mL of Hydrogen chloride (HCl) to the solution and stir for about 10 minutes.
- Sixth, adding 40 mL of phosphate buffer to the solution and stir for about 10 minutes.
- seventh, brought to a boil at 40-45°C for 5 hours by using lenses to collect sunlight heat then using Arduino system to control the temperature.
- Finally, collected the precipitation hyaluronic acid.



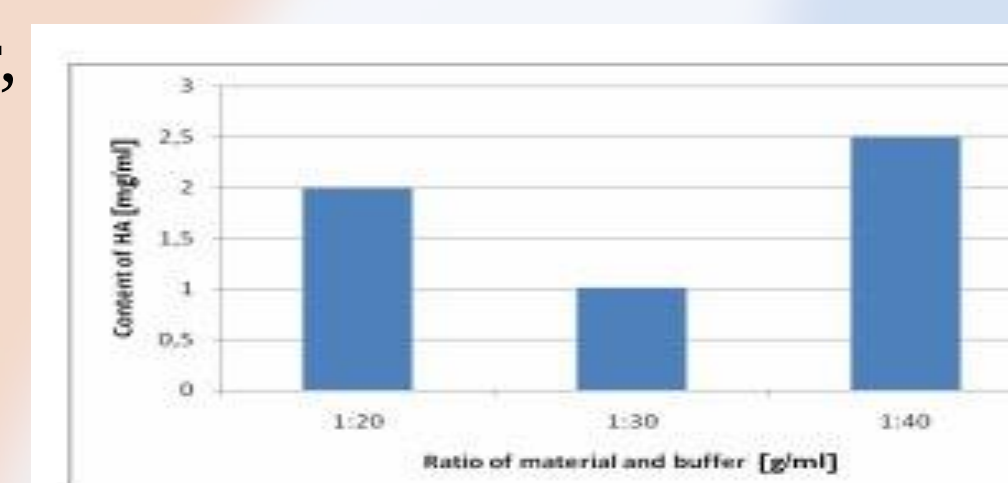
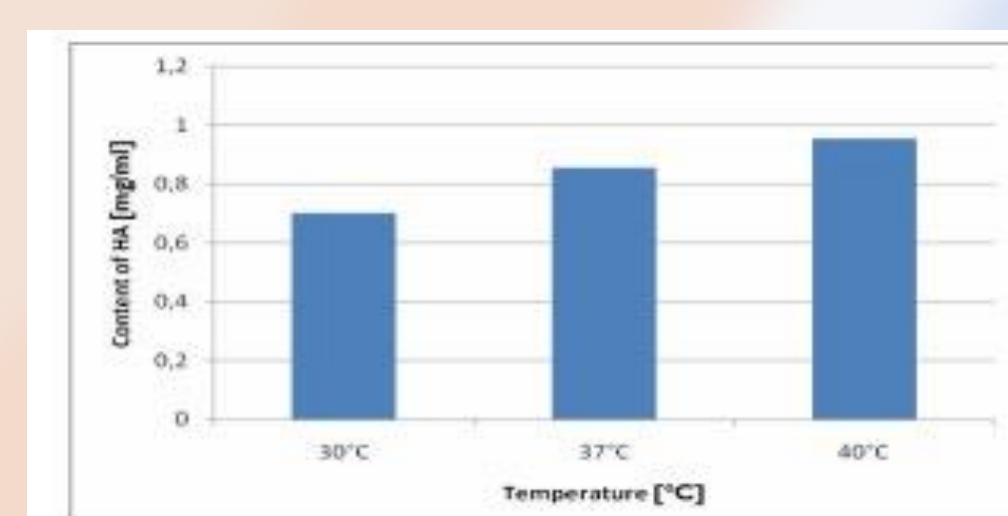
DATA & graph



Components	Concentration
Hyaluronic acid	9%
Amino acids	68--75%
collagen	10%
glucosamine	1%
Chondroitin sulfate	2%

Data analysis

How can we relieve the pain of patients with osteoarthritis? That was the question that led us to think about our idea. Where more than 10% in the world suffer from osteoarthritis, due to Obesity, Injury, and Heredity, and most doctors often resort to hyaluronic acid injections to relieve patient pain and to increase synovial fluid. But this acid is very expensive, with an estimated price of \$400-\$800. in addition to many patients require two or three syringes per treatment session, so we found that we can produce this acid from eggshells, total production of egg total population in world(76billion) where the average consumption is 161 eggs per person per year, and this is a very high rate. Therefore, we made steps to produce hyaluronic acid. First, dissolve the calcium carbonate of the shell and the membranes are recovered then dry it at room temperature or in the oven. After that, we put 100mL of water, two tables of digestin and 4mL of HCl in the beaker and stir it until the tables completely dissolved then grind 1g from the eggshell's membrane and added it to the solution then boil at 40-45 degrees Celsius, and finally added phosphate buffer to precipitate the hyaluronic acid . So, we produced about 5% of hyaluronic acid per 1 gram of eggshell, which present about 1.027 mg as the human body contains about 15 grams of hyaluronic acid. As the amount of dose that a person gets when he suffers from roughness of the joints, it depends on the part that is being treated, but it should not be more than 1.5ml per treatment.



Results

Results are the end of any search, so often each of us searches for satisfactory results . so we do three trails

Trails	Amounts of hyaluronic acid
Trail 1	4.112% hyaluronic acid per 1 g of eggshells.
Trail 2	4.651 % hyaluronic acid per 1 g of eggshells.
Trail 3	5 % hyaluronic acid per 1 g of eggshells.

And we can say that we achieved our design requirements as its safety , have low cost and extracting hyaluronic acid from eggshell.

Conclusion

Osteoarthritis is a disease that hundreds of people suffer from, so thinking about relieving their pain instead of surgical operations was our goal, and after a lot of research and deep thinking and consulting doctors from different universities, we can say that we reach the solution which is producing hyaluronic acid as it is injected into a patient with osteoarthritis to get rid of the pain. Our idea is different as we depend on using unused resources, which as eggshells membrane and by doing physical and chemical processes (boiling and chemical reactions)We can produce hyaluronic acid and get rid of osteoarthritis pain.

Applications

Every research has a benefit. Research in general is only a means to reach a solution to a specific problem. Our goal is to relieve the pain of patients who suffer from osteoarthritis. Where the benefits of our project to man are :

- Pain relief for patients with osteoporosis.
- Produce hyaluronic acid in a different way instead of making it chemically.
- Exploitation of eggshells, where it is permanently wasted, which causes its fermentation to produce greenhouse gases.

And we can apply it on the ground in the chemistry research laboratory, so it is a simple process that we can perform ourselves by adding pepsin , HCl and phosphate buffer to the eggshell membrane at a certain temperature with a certain concentrations, then to collect the hyaluronic after precipitation.

Research Sources

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